

Max Heap Insertion

Insertion	Score	Heap Arrangement	Comparisons
1	78	[78]	0
2	92	[92, 78]	1
3	65	[92, 78, 65]	1
4	88	[92, 88, 65, 78]	2
5	95	[95, 92, 65, 78, 88]	2
6	72	[95, 92, 72, 78, 88, 65]	2
7	84	[95, 92, 84, 78, 88, 65, 72]	2
8	90	[95, 92, 84, 90, 88, 65, 72, 78]	2

Total Comparisons During Insertion = **12**

Final Max Heap: **[95, 92, 84, 90, 88, 65, 72, 78]**

Highest Score = **95**

Linear Search Trace Table

Step	Current Score	Current Maximum	Comparison
Initial	78	78	—
1	92	92	92 > 78
2	65	92	65 > 92 → No
3	88	92	88 > 92 → No
4	95	95	95 > 92
5	72	95	72 > 95 → No
6	84	95	84 > 95 → No
7	90	95	90 > 95 → No

Total Linear Search Comparisons = **7**

Highest Score = **95**

Complexity Analysis

Operation	Max Heap	Linear Search
Find Maximum	$O(1)$	$O(n)$
Insert New Score	$O(\log n)$	$O(1)$ if appended
Repeated Maximum Queries	Efficient	Requires scanning
Space Complexity	$O(n)$	$O(n)$ for input array

Performance Comparison

Feature	Max Heap	Linear Search
Highest Score	95	95
Find Maximum	$O(1)$	$O(n)$
Insert Score	$O(\log n)$	$O(1)$
One-time Search	Less suitable	More suitable
Continuous Maintenance	More suitable	Less suitable

Analysis and Justification

Linear Search is simple and suitable when the highest score needs to be found only once from a small list.

Max Heap is preferable when scores are continuously added and the highest score needs to be accessed repeatedly. In a Max Heap, the maximum score is always at the root, so it can be found in $O(1)$ time, while insertion takes $O(\log n)$ time.

As the number of students increases, repeated Linear Search becomes less efficient.

Conclusion

The highest score obtained using both Max Heap and Linear Search is **95**.

For a one-time search, Linear Search is simpler and performs only **7 comparisons** for the given data.

For continuously maintaining and querying the highest score, **Max Heap is more suitable** because it provides **$O(1)$ maximum access** and **$O(\log n)$ insertion**.

Therefore, **Max Heap is recommended** for continuous score management.